

COMP702
MSc Project
Ethical Conduct

Thanks to: Dr Stuart Thomason

Collective Ethical Approval

- Ethical conduct includes personal, professional and academic integrity
- The university has strict procedures in place for any projects that involve **human subjects, animals, human tissue, or personal data**
 - If you want to work with any of these you must submit an ethics application
 - This can be complex and takes time to be approved (impractical for student projects)
- We have obtained a **collective ethical approval** for projects that...
 - Only use humans in requirements analysis and software evaluation tasks
 - Make use of data sets that are already in the public domain (including human/animal)
 - Analyse anonymous data from public sources or APIs (eg. social media posts)
- **You cannot conduct other surveys, experiments or activities with humans**

Data and Participant Categories

- Data categories...
 - A – No use of data derived from humans or animals
 - B – Anonymous data about humans or animals from open sources or public APIs
 - C – Anonymous data about humans extracted from social media via public APIs
 - D – Anonymous medical data about humans from open sources
 - E – Published data sets involving identifiable humans from open sources
- Participant categories...
 - 0 – No use of human participants in any activity
 - 1 – Human participants used during requirements analysis
 - 2 – Human participants used during software evaluation
 - 3 – Human participants used during requirements analysis and software evaluation

Ethical Checkpoints

- Our collective approval is based on commitments made in our application
 - We must clearly explain the ethical guidance to you
 - You must explicitly confirm that you have read and followed the guidance
 - We must have a process to check that you understand the guidance and followed it
- Each assessment acts as a checkpoint
 - You must include a statement of ethical compliance
 - There is a marking criterion that asks each supervisor to check the ethical content
- Failure to follow the ethical guidance constitutes academic misconduct
 - Will be processed as a Category E offence (the most serious category)
 - Could result in termination of studies in extreme cases
 - Usually results in a zero mark for the project module and suspension of studies

Statement of Ethical Compliance

- Towards the start of every assessment you **must** include a clear statement
 - State the data and participant categories that your project falls into
 - Confirm that you have followed (or will follow) the ethical guidelines

Statement of Ethical Compliance

Data Category: B

Participant Category: 2

I confirm that I have read the ethical guidelines and will follow them during this project.

- Add more detail in the written parts of the assessments
 - CA1: Data Sources; Project Ethics & Human Participants
 - CA3: Requirements Analysis; Evaluation; Project Ethics

Requirements Analysis & Software Evaluation

- Requirements analysis involves speaking to other people to find out...
 - What they currently do (eg. if you are trying to automate a manual process)
 - What they want you to implement (eg. if they are acting as a client for your project)
- Evaluation involves showing your software to other people to get feedback
 - Comments on errors or problems encountered during use, usability, navigability, etc.
 - Measurements or observations about speed, accuracy, and so on
- You might not need input from other people
 - Almost all students come up with a list of requirements on their own
 - You can test and use the software yourself (with dummy users if necessary)
- But if you do involve other people you must obtain **informed consent**

Permitted Activities

- Interviews – Asking people what they need or what they think of the result
- Discussions – Similar to interviews, but with a more relaxed and free style
- Observations – Watching people use software or carry out a manual process
- Narrations – Similar to observations, but asking them to speak aloud as well
- Questionnaires – Giving people a fixed set of questions (online or on paper)
- You can make audio and screen recordings of participants with their consent
 - Transcribe and anonymise audio as soon as possible afterwards
 - Delete the recordings as soon as you have gathered the information you need
- You must not carry out any other kind of activity or experiment with humans (or animals)

Participant Information & Consent

- The participant information sheet covers standard questions that people might have before they start
 - How their data is collected, stored and used
 - How they can withdraw and request their data to be deleted
- The participant consent form allows each person to confirm that they are happy to take part
 - Makes it clear that the project is part of the University of Liverpool
 - Provides contact information for the student, supervisor and project coordinator
- Fill in the top part of the consent form with your project details
 - Give one to each participant and ask them to initial the box if they agree to take part
 - Give completed forms to your supervisor (either electronic or printed)

Protecting Participant Data

- Participant data is anything that you collect from another person as part of your software evaluation
 - Should only be collected with the permission of participants
 - Should be stored securely on a password-protected computer or university file store
 - Should not be shared with anyone else
 - Should be deleted after graduation
 - Should not contain any personal info about participants or reveal their identity
- Your dissertation should explain how your evaluation was carried out
 - All data should be aggregated before it appears in your video or dissertation
 - Scores, results, comments and analysis should be presented in summary form
 - Make sure everything is fully anonymised
 - Don't reveal anyone's identity by mistake (ie. from the comments they made)

Using Dummy Data

- Most evaluation tasks will require people to enter some data
 - Maybe a registration form for a website or app
 - Personal data to test the features and functionality of the software (eg. age, weight)
- Advise participants to **always enter false (dummy) data**
 - They can enter false age/weight/etc without affecting their use of the software
 - You should test boundary cases (extremes of data) before the evaluation phase
 - Test registration or 'sign up' forms with pretend users
 - **Do not collect real names, addresses, location data, passwords, or any other PII**
- Their focus is to provide feedback on what they think of your software
 - They can do this without needing to provide real data
 - Dummy data should be realistic but not revealing

Contacts

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- Please read the full guidance on Canvas before commencing your project